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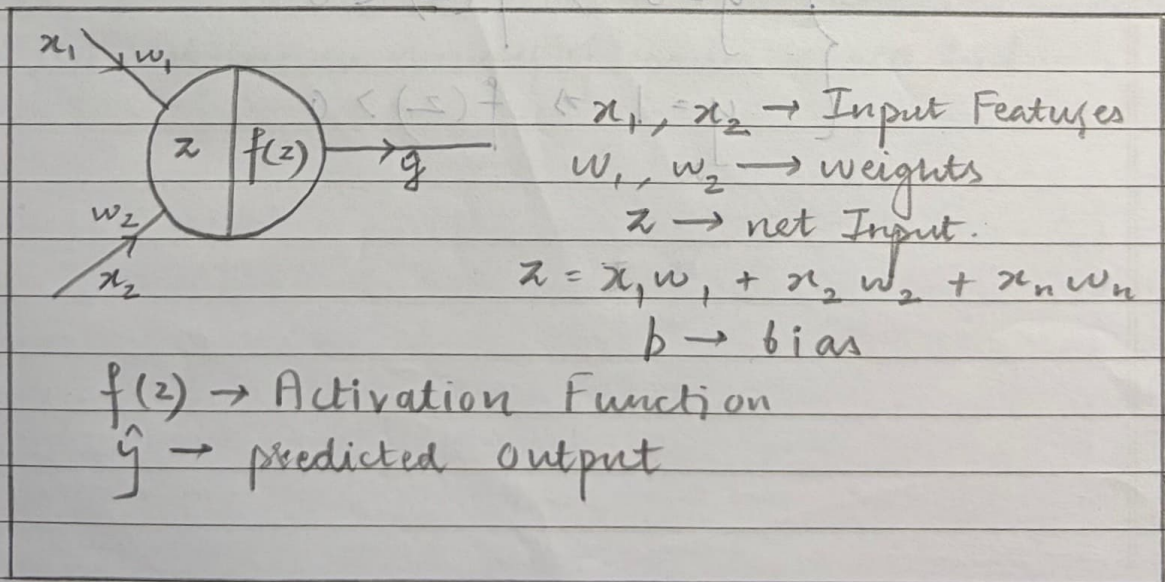
Div: B Subject: Deep Learning.

1) What is a Perceptron?

Ans: A Perceptron is the simplest type of artificial neural network used for binary classification. A single perceptron can classify only linearly separable data. It consists of input values, weights, a bias, and an activation function that together determine the output. The perceptron learns by adjusting its weights based on errors during training. Nonlinearly separable data requires a multi-layer perceptron with hidden layers and nonlinear activation functions.

2) Architecture of a Perceptron?

Ans:-



2) Mathematical model

The mathematical model of a perceptron considers the weighted sum of the inputs and adds a bias. The result is then passed through an activation function to produce the final output. (predicted)

$$y = f\left(\sum_{i=1}^n w_i x_i + b\right); \quad z = \sum_{i=1}^n w_i x_i + b$$

Activation functions is decided by the programmer eg:- step-function, Sigmoid, ReLU, Tanh

x_i = input values; w_i = weights; b = Bias; y = Output
 f = Activation function.

(1958)
Step Function by Frank Rosenblatt is the most basic ~~for~~ Activation Function.

$$\begin{cases} y = 0 \Rightarrow f(z) \leq 0 \\ y = 1 \Rightarrow f(z) > 0 \end{cases}$$

4) Learning Rule

Ans:- The perceptron Learning Rule updates the weights and bias based on the prediction error. If the predicted output is incorrect, the weights and bias are adjusted to reduce the error.

Weight Update:-

$$w_{\text{new}} = w_{\text{old}} + \eta (y_{\text{actual}} - y_{\text{predicted}}) x$$

Bias Update

$$b_{\text{new}} = b_{\text{old}} + \eta (y_{\text{actual}} - y_{\text{predicted}})$$

where:-

η (eta) $\Rightarrow$ Learning Rate ; y_{actual} $\Rightarrow$ Actual output value;
 x $\Rightarrow$ Input value (vector) ; $y_{\text{predicted}}$ $\Rightarrow$ Predicted output value.

$\Rightarrow$ The training data is processed repeatedly over multiple epochs. In each epoch, the perceptron updates its weights and bias until the classification error is minimised or the maximum number of epochs is reached.

5) Algorithm of a Single Perceptron

Ans:-

Step 1 $\Rightarrow$ Initialize the weights. Import the required Library (Numpy) for Numerical Computations.

Step 2 $\Rightarrow$ Store the input features and target output in separate variables. Initialize the weights and bias with small values or zeros.

Step 3 $\Rightarrow$ Define activation function (step function) to produce binary output (0, 1). Set learning rate (η) and no. of epochs.

Step 4 $\Rightarrow$ For each epoch:- $f(z) \Rightarrow z = \sum_{i=1}^n w_i x_i + b$

Step 5 $\Rightarrow$ Compute the error: Actual value - Predicted value.

Step 6 $\Rightarrow$ Update the weights and bias using Perceptron Learning Rule.

Step 7:- Repeat step 4, 5, 6 until the error becomes zero or the maximum number of epochs is reached.

Step 8:- Display the final weight, bias, and predictions for the given input data.